Geographic and Temporal Variation in Medicare Part D Formulary Restrictiveness for Diabetes Medications, 2020-2025

Capstone Deliverable: Quantitative Analysis with GIS / Interactive Website

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Abstract

Background: The Inflation Reduction Act of 2022 restructured Medicare Part D plan liability, creating new financial incentives for plans to restrict access to high-cost medications. This study examined geographic and temporal variation in formulary restrictiveness for four diabetes medication classes during this policy transition. **Methods:** Using publicly available Centers for Medicare & Medicaid Services (CMS) Part D Formulary Files, plan information files, and enrollment data from 2020 to 2025, this descriptive study calculated enrollment-weighted utilization management rates and composite restrictiveness scores for GLP-1 receptor agonists, insulin, SGLT2 inhibitors, and DPP-4 inhibitors across approximately 3,000 U.S. counties. Geographic variation was assessed through state-level comparisons and county-level choropleth mapping using ArcGIS Pro. **Results:** GLP-1 receptor agonist prior authorization rates increased from 6.7% to 100.0% of enrollment-weighted plans between 2020 and 2025, while insulin (2.5% to 1.2%), SGLT2 inhibitor (0.6% to 0.1%), and DPP-4 inhibitor (0.2% to 0.1%) rates remained low. The GLP-1 composite restrictiveness score rose from 1.08 to 2.03 on a 0–4 scale. State-level GLP-1 prior authorization rates in 2024 ranged from 57% to 100%, and county-level mapping revealed within-state heterogeneity and a Southeast cluster of higher overall restrictiveness. **Conclusion:** Medicare Part D plans dramatically and selectively increased GLP-1 formulary restrictions during the IRA implementation period, with significant geographic variation in the pace of adoption. Continued monitoring and potential policy interventions are warranted to ensure access to this clinically important medication class.

Background

Type 2 diabetes affects approximately 24.4% of Americans aged 65 and older, representing over 14 million Medicare beneficiaries and making it one of the most prevalent chronic conditions in the Medicare population.[1] Effective diabetes management requires consistent access to prescription medications, yet Medicare Part D formularies (the lists of covered drugs and their associated access restrictions) have become increasingly restrictive over time. Between 2011 and 2020, prior authorization, step therapy, and quantity limits expanded dramatically, with restricted compounds increasing from 31.9% to 44.4% across Part D plans.[2] Most notably, prior authorization requirements for GLP-1s surged from approximately 5% of plans in the early 2020s to nearly 100% by 2025.[3,4] These formulary restrictions create substantial barriers to medication access for millions of beneficiaries and may disproportionately affect vulnerable populations including racial and ethnic minorities, lower-income beneficiaries, and those with limited health literacy who face greater challenges navigating administrative barriers.[1]

The period from 2020 to 2025 represents a particularly dynamic era for Part D policy, marked by multiple major reforms aimed at improving medication affordability and access. The Part D Senior Savings Model (2021-2023) allowed plans to voluntarily cap insulin copayments at $35 per month, demonstrating improved adherence in participating plans.[5] The Inflation Reduction Act of 2022 expanded this insulin cap program-wide beginning in January 2023 and introduced additional cost-sharing protections, including a $2,000 annual out-of-pocket spending cap effective January 2025.[6] However, these same policy changes fundamentally restructured Part D financial liability. Under the benefit redesign, plans' share of costs for high-spending beneficiaries in the catastrophic coverage phase—the stage reached after significant annual drug spending—increased from 15% to 60% for brand-name drugs. This shift created strong financial incentives for plans to adopt more aggressive utilization management strategies, particularly prior authorization requirements, for expensive medications likely to push beneficiaries into that spending phase.[6]

Against this policy backdrop, existing research establishes that formulary restrictions impact medication access and utilization patterns. Studies consistently demonstrate that prior authorization requirements, step therapy protocols, and higher cost-sharing tiers reduce rates of medication initiation and continuation.[7] Variation in access to diabetes medications is documented across regions, with

initiation rates for newer glucose-lowering drugs varying fivefold across U.S. hospital referral regions.[1] Additionally, substantial racial and socioeconomic disparities exist in medication access, with Black beneficiaries experiencing lower rates of SGLT2 inhibitor and GLP-1 receptor agonist initiation even after adjusting for clinical factors.[1,5]

Despite these dramatic policy shifts, comprehensive characterization of how formulary restrictiveness varies across counties and over time remains limited and underexplored. For example, although national trends in formulary restrictions have been documented for the 2011-2020 period,[2] comprehensive characterization of county-level variation and disparities in formulary restrictiveness remains lacking. Additionally, despite the dramatic formulary changes occurring during 2020-2025 (particularly the near-universal adoption of prior authorization for GLP-1 receptor agonists), temporal trends during this critical policy transition remain poorly documented. Moreover, existing approaches for measuring formulary restrictiveness vary widely across studies,[8] limiting comparability and hindering development of validated metrics that can be applied consistently in future research and policy monitoring.

This project addresses these gaps by systematically characterizing county-level and temporal variation in Medicare Part D formulary restrictiveness for diabetes medications from 2020 through 2025. By documenting the extent and distribution of this variation, findings will identify markets where restrictive formularies may particularly affect vulnerable populations. The temporal analysis will reveal how plans responded to major policy changes, informing ongoing debates about balancing medication cost control with ensuring access to effective therapies. This project will establish validated measures of restrictiveness using publicly accessible data sources, providing a replicable framework for continued monitoring of Part D formulary policies.

Methods

Study Design

This descriptive study uses secondary analysis of publicly available Centers for Medicare & Medicaid Services (CMS) administrative data to characterize geographic and temporal variation in Medicare Part D formulary restrictiveness for diabetes medications. The analysis examines all Part D

plans, both standalone Prescription Drug Plans (PDPs) and Medicare Advantage Prescription Drug Plans (MA-PDs), from plan years 2020 through 2025. This period encompasses the Part D Senior Savings Model (2021-2023), the Inflation Reduction Act insulin copay cap implementation (January 2023), and the benefit redesign effective January 2025.

Data Sources

Three publicly available CMS data sources were used. First, the Part D Formulary Files contain plan-level information on covered medications including tier placement, prior authorization requirements, step therapy protocols, and quantity limits for each National Drug Code (NDC). These files are released annually and reflect formulary status as of each plan year. Second, the Part D Plan Characteristics Files provide plan-level identifiers, enabling linkage between formulary data and enrollment information, including contract and plan identifiers, plan names, and service area counties. Third, the Part D Enrollment Files provide county-level monthly enrollment counts by contract and plan, enabling calculation of enrollment-weighted restrictiveness measures. All data are publicly available through the CMS website (data.cms.gov) at no cost, eliminating data use agreement requirements and enabling replication. This reliance on freely accessible public data represents a methodological contribution, demonstrating that policy-relevant formulary research can be conducted without costly data use agreements or lengthy approval processes.

Study Population and Sample

The unit of analysis is the plan-county-year, representing each Part D plan's formulary characteristics within each county it serves for each study year. All Part D plans offering prescription drug coverage during 2020–2025 were included, encompassing both standalone Prescription Drug Plans and Medicare Advantage Prescription Drug Plans. Plans were excluded if they had missing or incomplete formulary data for diabetes medication classes of interest or had zero enrollment in a given county-year.[3,9]

Measures

Diabetes Medication Classes. Four major diabetes medication classes were examined, consistent with prior formulary research on glucose-lowering therapies[1,7]: (1) insulin products, including rapid-acting, short-acting, intermediate-acting, long-acting, mixed, and concentrated formulations; (2) GLP-1 receptor agonists, including injectable semaglutide (Ozempic), oral semaglutide (Rybelsus), tirzepatide (Mounjaro), dulaglutide (Trulicity), and liraglutide (Victoza); (3) SGLT2 inhibitors, including empagliflozin (Jardiance), dapagliflozin (Farxiga), and canagliflozin (Invokana); and (4) DPP-4 inhibitors, including sitagliptin (Januvia), linagliptin (Tradjenta), saxagliptin (Onglyza), and alogliptin (Nesina). Medications were identified in the CMS formulary files using RxNorm Concept Unique Identifiers (RXCUIs) verified against the National Institutes of Health RxNav database, and linked to National Drug Codes using the CMS Formulary Reference File crosswalks.

Formulary Restrictiveness Measures. Four dimensions of formulary restrictiveness were measured for each medication class within each plan, drawing on established approaches in Part D formulary research.[2,8] Prior authorization prevalence was calculated as the percentage of medications within each class requiring prior authorization for coverage. Step therapy prevalence represents the percentage of medications requiring trial and failure of alternative agents before coverage. Quantity limits are the percentage of medications subject to dispensing quantity restrictions. Tier placement categorizes medications as preferred (Tiers 1-3) versus non-preferred (Tiers 4-5) or specialty tier, following the tier classification approach used by Buttorff et al.[9] in their analysis of insulin formulary trends.

A composite restrictiveness index was constructed by assigning one point for each restriction type applied to a medication (prior authorization, step therapy, quantity limits, non-preferred or specialty tier placement), yielding medication-level scores ranging from 0 (fully unrestricted) to 4 (maximally restricted). Class-level restrictiveness was calculated as the mean score across all medications within each therapeutic class. This approach adapts the methodology of Shen et al.,[8] who developed a three-level restriction classification for oral diabetes medications. Following their framework, each medication will also be categorized as: (1) unrestricted (covered without prior authorization, step therapy, or non-preferred tier placement); (2) partially restricted (subject to one or two restriction types); or (3) fully restricted (subject to three or more restriction types or formulary exclusion). This categorical

classification enables comparison with prior research while the continuous index provides greater granularity for detecting variation.

Geographic Variables. County-level formulary restrictiveness was calculated as the enrollment-weighted average of plan-level restrictiveness measures, with weights proportional to each plan's enrollment within the county. This enrollment-weighting approach, consistent with methods used by Liu et al.[4] in their quarterly analysis of GLP-1 coverage, reflects the formulary environment experienced by the typical Medicare beneficiary in each county rather than giving equal weight to plans regardless of market share. County-level SSA codes from CMS geographic locator files were converted to Census FIPS codes using the National Bureau of Economic Research SSA-to-FIPS crosswalk to enable geographic mapping and visualization.[1]

Analysis

All data processing and statistical analyses were conducted using Python (version 3.x) with the pandas library for data manipulation. Custom scripts filtered raw formulary files to relevant diabetes medications using verified RXCUI codes, calculated utilization management rates and composite restrictiveness scores by drug class and plan, and generated enrollment-weighted county-level measures by merging formulary data with plan service area and enrollment files. Descriptive statistics characterized the distribution of formulary restrictiveness measures across drug classes, counties, and years. Frequencies and percentages described the prevalence of prior authorization, step therapy, and quantity limit requirements. Temporal trends were examined by calculating annual changes in enrollment-weighted utilization management rates across the study period. Key policy implementation dates—January 2023 (IRA insulin cap) and January 2025 (benefit redesign)—were annotated on trend figures to assess whether observable inflection points in formulary restrictiveness coincided with policy changes, as has been documented for GLP-1 receptor agonists in recent research.[3,4]

Geographic variation was assessed through county-level mapping of enrollment-weighted restrictiveness measures. County-level choropleth maps were created using ArcGIS Pro to display the distribution of GLP-1 prior authorization rates and average formulary restrictiveness scores across the continental United States. State-level comparisons were generated to examine between-state variation in GLP-1 prior authorization adoption during the transition period. This study qualifies for exemption from

Institutional Review Board review under 45 CFR 46.104(d)(4) as research involving secondary analysis of publicly available, aggregate administrative data containing no individually identifiable information.

Results

Analytic Sample

The analytic sample included formulary entries for four diabetes medication classes across 2020 to 2025, with the number of unique Part D formularies declining from 457 in 2021 to 383 in 2025 as the plan market consolidated. The number of counties with enrollment-weighted data ranged from 2,972 in 2020 to 3,154 in 2025, spanning 50 states, the District of Columbia, and Puerto Rico. Alaska was excluded from all analyses due to incomplete county-code matching in the SSA-to-FIPS crosswalk; geographic visualizations display the continental United States. GLP-1 receptor agonist formulary entries increased substantially over the study period, from 1,561 in 2020 to 6,016 in 2025, reflecting the expanding market for this drug class. Insulin entries were the most numerous in each year, ranging from 9,932 to 11,908. SGLT2 inhibitor and DPP-4 inhibitor entries remained relatively stable, each with approximately 1,600 to 2,000 entries per year. Full sample characteristics are presented in Appendix Table A1.

Temporal Trends in Utilization Management

The most notable temporal trend was the dramatic increase in prior authorization requirements for GLP-1 receptor agonists. Enrollment-weighted GLP-1 prior authorization rates remained stable at approximately 6% to 7% from 2020 through 2023, then surged to 78.1% in 2024 and reached 100.0% in 2025—a 93-percentage-point increase over the study period (Figure 1). In contrast, prior authorization rates for the other three drug classes remained low and stable throughout the study period: insulin ranged from 1.2% to 2.5%, DPP-4 inhibitors from 0.1% to 0.3%, and SGLT2 inhibitors from 0.1% to 0.6%.

Examination of all three utilization management tools revealed that the GLP-1 surge was specific to prior authorization (Figure 2). GLP-1 step therapy requirements, which had been approximately 8% in 2020, declined to near zero by 2025 as plans shifted to prior authorization. Quantity limits were consistently high across all four drug classes (approximately 85% to 99% of formulary entries) throughout

the study period, indicating that quantity limits function as a baseline utilization management tool rather than a targeted restriction.

Figure 1. Prior Authorization Rates for Diabetes Medications in Medicare Part D Plans, 2020–2025

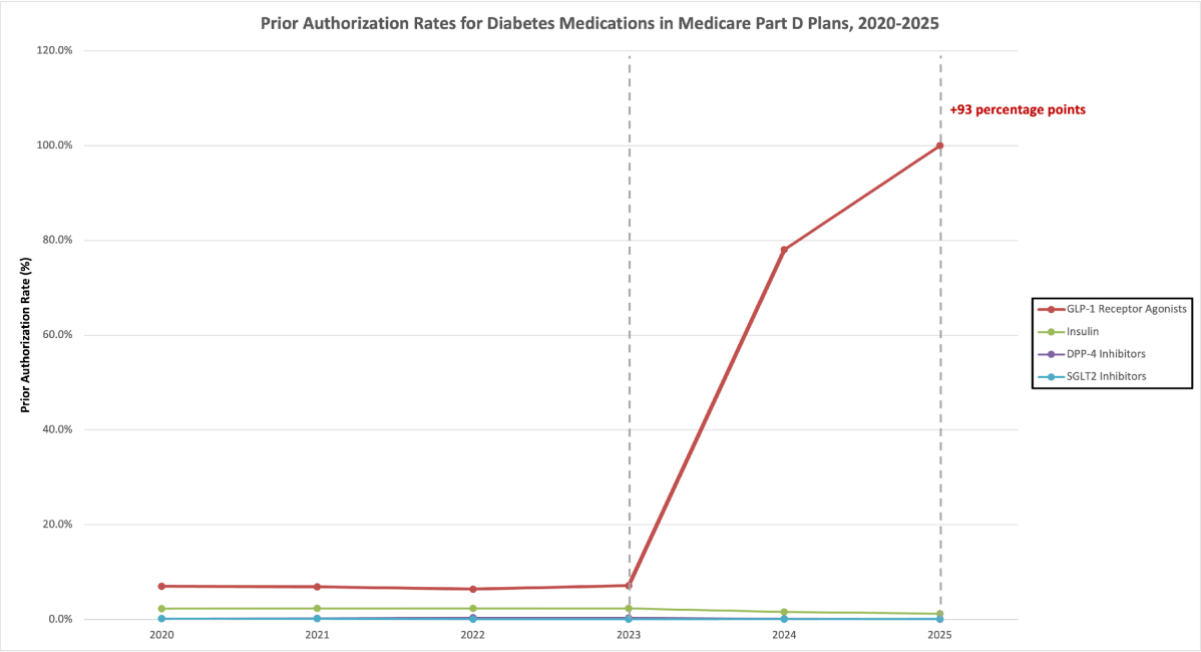
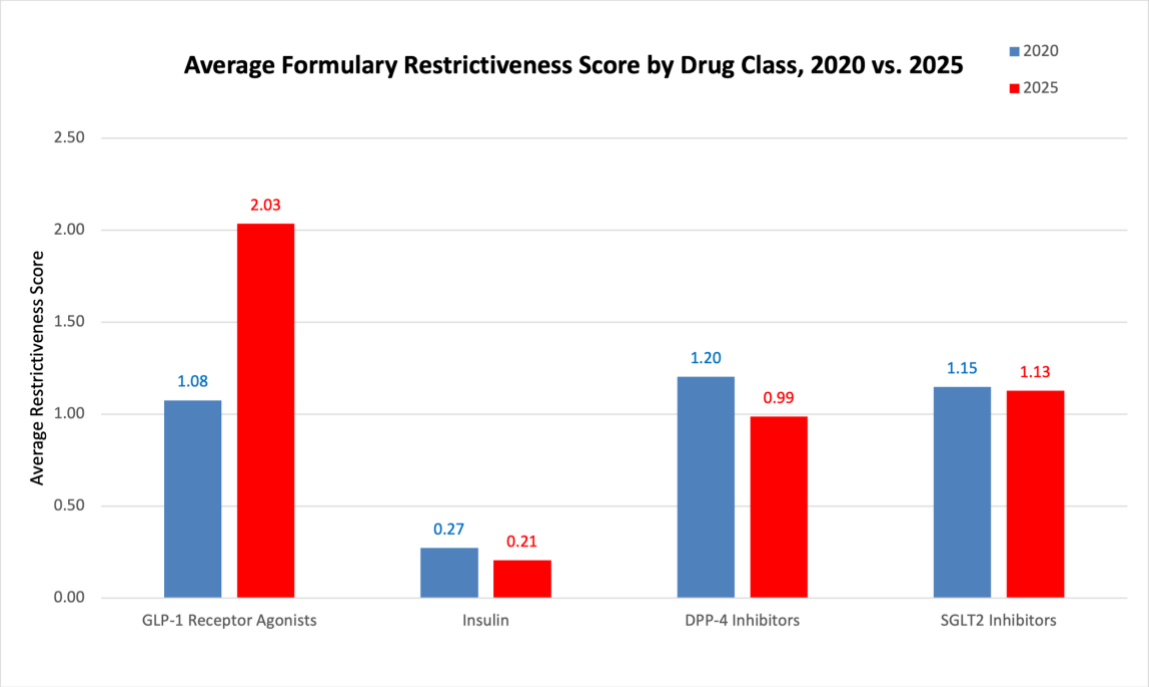


Figure 2. Utilization Management Tools by Diabetes Medication Class, Medicare Part D, 2020–2025

Overall Formulary Restrictiveness

The composite restrictiveness score, which combines prior authorization, step therapy, quantity limits, and high-tier placement into a single 0–4 scale, captured the divergence between drug classes over time (Figure 3). GLP-1 receptor agonists saw the largest increase, rising from a mean score of 1.08 in 2020 to 2.03 in 2025. Insulin remained the least restricted class, with scores declining slightly from 0.27 to 0.21, consistent with increased protections under the IRA's $35 copay cap. DPP-4 inhibitor restrictiveness decreased modestly from 1.20 to 0.99, and SGLT2 inhibitor scores were essentially unchanged at 1.15 to 1.13.

Figure 3. Average Formulary Restrictiveness Score by Drug Class, 2020 vs. 2025



Geographic Variation

Substantial geographic variation in GLP-1 prior authorization rates was observed across states, particularly in 2024—the year of greatest between-state heterogeneity before convergence to near-universal requirements in 2025 (Figure 4). State-level enrollment-weighted GLP-1 prior authorization rates in 2024 ranged from approximately 57% in Montana to 100% in Massachusetts. New England states generally adopted prior authorization requirements earliest, while Mountain West and Southeast states

were slower to implement these restrictions. By 2025, this variation had largely collapsed, with all states reaching near-complete GLP-1 prior authorization coverage.

County-level choropleth mapping revealed within-state heterogeneity that state-level averages obscure (Figures 5 and 6). The GLP-1 prior authorization map for 2024 shows counties ranging from less than 20% to above 80% even within the same state, reflecting variation in the mix of Part D plans available to beneficiaries in different counties. The average formulary restrictiveness score map reveals a distinct regional pattern, with a cluster of higher overall restrictiveness in the Southeast that is not fully explained by the GLP-1 prior authorization pattern alone—suggesting that plans in these regions apply more restrictive formulary strategies across multiple drug classes.

Figure 4. GLP-1 Prior Authorization Rate by State, 2020 vs. 2024

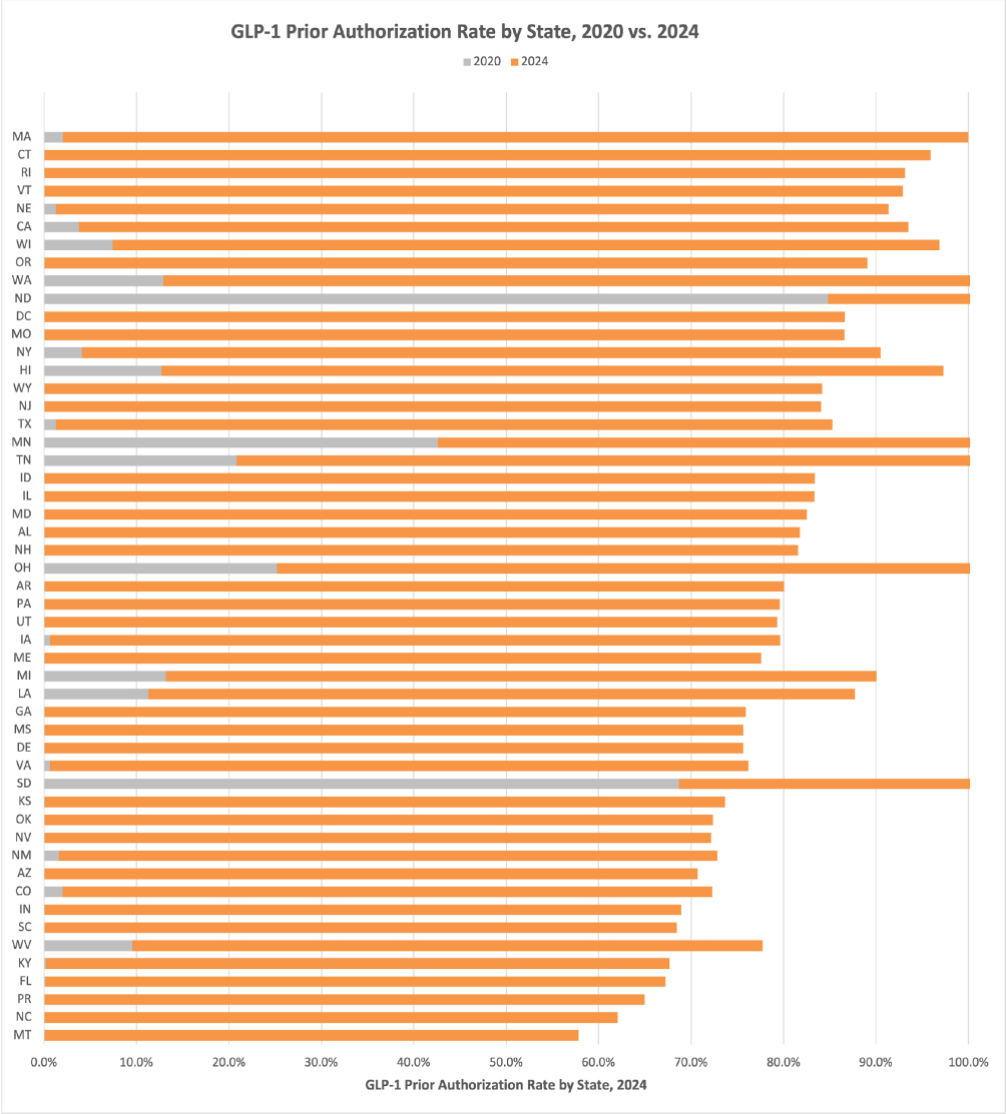


Figure 5. County-Level GLP-1 Prior Authorization Rates, 2024

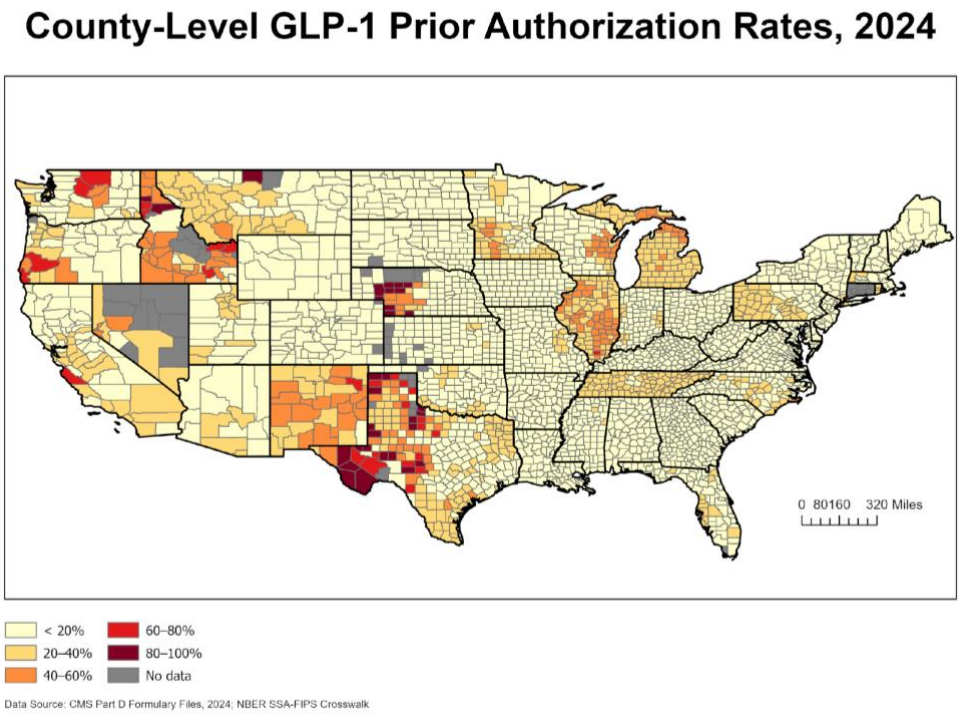
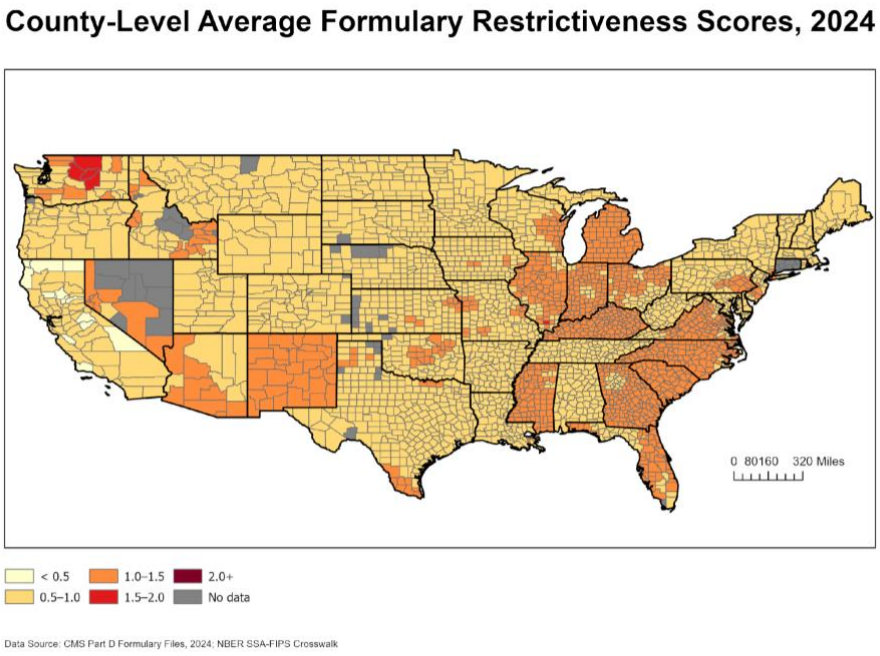


Figure 6. County-Level Average Formulary Restrictiveness Scores, 2024



Discussion

This study documented a dramatic and targeted increase in GLP-1 receptor agonist prior authorization requirements in Medicare Part D formularies between 2020 and 2025, while restrictions on insulin, SGLT2 inhibitors, and DPP-4 inhibitors remained essentially unchanged. The enrollment-weighted GLP-1 prior authorization rate rose from 6.7% to 100.0% over this period, with the sharpest acceleration occurring between 2023 and 2025. Geographic analysis revealed substantial county-level variation during the transition period, with a distinct regional pattern of higher overall formulary restrictiveness in the Southeast.

The timing of the GLP-1 prior authorization surge is consistent with plan responses to the IRA's benefit redesign, which increased plan liability in the catastrophic coverage phase from 15% to 60% for brand-name drugs effective January 2025.[6] GLP-1 receptor agonists, with monthly costs often exceeding $1,000, frequently push beneficiaries into the catastrophic phase—making these medications a primary target for cost containment.[3] That plans selectively increased restrictions on GLP-1s while leaving less expensive drug classes largely untouched suggests a financially motivated response to the new liability structure rather than a broad shift in formulary management philosophy.

These findings are consistent with recent research documenting the GLP-1 prior authorization surge. Klebanoff et al. reported that the share of Part D plans requiring prior authorization for GLP-1s increased from approximately 5% to nearly 100% between 2020 and 2025,[3] and Liu et al. found similar trends specifically for semaglutide and tirzepatide.[4] This study extends prior work in two ways: first, by documenting county-level geographic variation that national and plan-level analyses do not capture; and second, by comparing four diabetes drug classes simultaneously, demonstrating that the restriction increase was isolated to GLP-1s. The stability of insulin restrictions aligns with Buttorff et al.'s finding that the IRA's $35 copay cap shaped insulin formulary strategies,[9] and the modest decline in DPP-4 restrictiveness may reflect these older medications facing less cost-driven scrutiny as GLP-1s absorbed plan attention.

The geographic variation observed in this study warrants particular attention. The staggered adoption of GLP-1 prior authorization across states—with New England states implementing restrictions earlier and Mountain West and Southeast states adopting them later—suggests that plan responses to

anticipated IRA changes were not uniform. The Southeast cluster of higher overall formulary restrictiveness visible in the average score map may reflect regional differences in plan market structure, Medicare Advantage penetration, or pharmacy benefit manager contracting practices. These findings underscore the value of county-level analysis: beneficiaries' experiences with formulary barriers depend not just on national policy but on which plans are available in their specific county.

The geographic concentration of higher formulary restrictiveness in the Southeast is particularly concerning when considered alongside the region's demographic and health profile. The Southeast consistently ranks among the highest in diabetes prevalence, social vulnerability, and racial and ethnic minority Medicare enrollment—the same populations for whom prior research has documented lower rates of GLP-1 and SGLT2 inhibitor initiation even after adjusting for clinical factors.[1] If areas with the highest formulary restrictiveness also have the greatest disease burden and the most socially vulnerable populations, formulary policy may function as a structural contributor to existing disparities rather than a neutral administrative process. A planned extension of this analysis will explore this question through geographic overlay of county-level restrictiveness scores with indicators of social vulnerability, diabetes prevalence, and racial composition to assess whether high-restrictiveness counties are disproportionately concentrated in disadvantaged communities.

These results have implications for Medicare formulary oversight and ongoing policy development. The near-universal adoption of GLP-1 prior authorization raises questions about whether these requirements function as appropriate clinical gatekeeping or as de facto access barriers for a medication class with demonstrated cardiovascular and renal benefits.[7] Policymakers may consider whether GLP-1 access protections, analogous to the IRA's insulin copay cap, are warranted. Additionally, the methodology demonstrated here—using entirely free, publicly available CMS data—offers a reproducible framework for ongoing formulary surveillance that researchers, advocacy organizations, and regulatory agencies can adopt to monitor access trends over time.

The near-universal adoption of prior authorization for GLP-1 receptor agonists likely reflects plans' responses to multiple concurrent pressures beyond the IRA benefit redesign alone, including supply constraints and broader cost concerns driven partly by surging demand for GLP-1 medications for off-label weight management outside the Medicare population. Notably, Medicare Part D covers GLP-1s exclusively for type 2 diabetes management; the weight loss formulations (semaglutide as Wegovy and

tirzepatide as Zepbound) are distinct products not covered under Part D. Yet Medicare beneficiaries with diabetes bear the formulary consequences of demand pressures originating largely in younger, commercially insured, or cash-pay populations seeking these medications for cosmetic purposes. The result is that beneficiaries with the strongest clinical indication for GLP-1 therapy—those with type 2 diabetes and established cardiovascular or renal risk—now face universal administrative barriers that were implemented, at least in part, in response to utilization trends outside their population.

Limitations

Several limitations should be acknowledged. First, formulary files capture plan policies as designed but cannot assess actual beneficiary experiences with prior authorization approvals, denials, or appeals; plans may require prior authorization but approve most requests, or conversely may have stringent criteria resulting in frequent denials. Second, the descriptive design documents associations between policy periods and formulary changes but cannot establish causation. Third, county-level aggregation may mask within-county variation in plan availability and beneficiary experiences. Fourth, enrollment weighting assumes beneficiaries select plans independent of formulary restrictiveness for specific medications, which may not hold for beneficiaries with established diabetes medication regimens who actively consider formulary coverage when choosing plans. Fifth, the SSA-to-FIPS county code crosswalk resulted in incomplete geographic matching for Alaska, which was excluded from all analyses, and geographic visualizations display only the continental United States. Sixth, the analysis cannot fully disentangle the role of off-label demand for GLP-1 medications for weight management, which contributed to supply constraints and heightened plan scrutiny of this drug class during the study period; the observed increase in prior authorization may reflect plans responding to both the IRA's financial restructuring and broader market dynamics simultaneously. Despite these limitations, this analysis provides comprehensive characterization of geographic and temporal variation in diabetes medication formulary restrictiveness during a critical policy transition period, establishing baseline measures for ongoing monitoring and future research.

Conclusion

Between 2020 and 2025, Medicare Part D formulary restrictions on GLP-1 receptor agonists increased from 6.7% to 100.0% of enrollment-weighted plans requiring prior authorization, while insulin, SGLT2 inhibitor, and DPP-4 inhibitor restrictions remained largely unchanged. This targeted increase, concentrated in the period surrounding the Inflation Reduction Act's benefit redesign, was accompanied by significant geographic variation at the county level during the transition period. Given that GLP-1 receptor agonists have demonstrated cardiovascular and renal benefits for the more than 14 million Medicare beneficiaries with diabetes, continued monitoring of formulary access barriers—and consideration of targeted policy interventions—such as prior authorization standards that mandate expedited review for diabetes-indicated prescriptions, or copay protections analogous to the IRA's insulin cap—remains an important priority for ensuring that formulary cost containment does not compromise access to clinically essential therapies.

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Appendix

Table A1. Analytic Sample Characteristics by Drug Class and Year, Medicare Part D Formulary Files, 2020–2025

Year	Drug Class	Formularies	Entries	PA (%)	ST (%)	QL (%)	Restrict. Score
2020	GLP-1	441	1,561	6.2	16.9	84.9	1.08
	Insulin	447	9,932	3.2	2.4	26.3	0.27
	SGLT2	447	1,842	0.7	23.5	81.2	1.15
	DPP-4	447	2,034	0.7	15.2	88.2	1.20
2021	GLP-1	457	3,699	5.4	16.5	87.6	—
	Insulin	457	11,256	3.3	2.3	23.6	—
	SGLT2	457	1,914	0.5	8.2	84.0	—
	DPP-4	457	2,028	0.9	12.1	89.5	—
2022	GLP-1	457	4,193	7.6	11.4	87.2	—
	Insulin	457	11,569	3.5	1.5	23.3	—
	SGLT2	457	1,956	0.4	3.1	83.9	—
	DPP-4	457	1,813	1.7	3.5	88.3	—
2023	GLP-1	450	5,552	15.5	16.5	88.3	—
	Insulin	450	11,908	3.7	1.8	21.9	—
	SGLT2	450	1,960	0.4	2.2	84.7	—
	DPP-4	450	1,771	1.8	3.5	93.9	—
2024	GLP-1	421	6,149	88.1	3.4	94.9	—
	Insulin	421	11,671	2.7	1.4	21.0	—
	SGLT2	421	1,804	0.2	0.7	84.3	—
	DPP-4	421	1,785	0.4	2.7	93.8	—
2025	GLP-1	383	6,016	98.9	0.3	97.8	2.03
	Insulin	383	10,428	2.6	0.1	31.3	0.21
	SGLT2	383	1,642	0.5	0.2	96.2	1.13
	DPP-4	383	1,680	0.3	2.2	96.1	0.99

Note. PA = prior authorization; ST = step therapy; QL = quantity limits. Percentages represent the proportion of formulary entries within each drug class subject to each utilization management requirement. Restrictiveness scores (0–4 scale) shown for 2020 and 2025 only; intermediate years available upon request. Entry counts reflect the number of unique plan-medication combinations in CMS Part D Formulary Files. Em dash (—) indicates data not displayed.